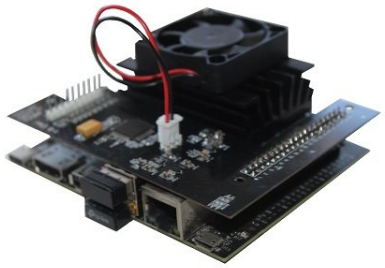
PRODUCT DATASHEET · SKU CQA240229JN

8-Channel EEG Shield for NVIDIA Jetson Nano

An open-source shield that turns the NVIDIA Jetson Nano into a brain-computer interface with a GPU on board. A Texas Instruments ADS1299 reads EEG, EMG and ECG at 24 bits; the Jetson reads it over SPI and can run neural networks on the signal in real time, right on the device.



8	24-bit	250 SPS	GPU	40-pin
Channels + REF + BIAS	TI ADS1299 front end	Default, up to 16 kSPS	128 CUDA cores on the Nano	Jetson header, SPI1

01 OVERVIEW

Key features

- **Plugs onto the Jetson Nano**: elevated 40-pin socket; L-shaped board clears the heatsink and fan.
- **Research-grade front end**: TI ADS1299, 8 simultaneous 24-bit channels, programmable gain 1–24.
- **Low-noise supply**: ultra-low-noise regulators on the shield; supply test points for checks.
- **Real-time ML**: run PyTorch or TensorRT models on the EEG stream on the Jetson GPU.
- **PiEEG Server**: dashboard, WebSocket, CSV, LSL and OSC with one flag, `--device jneeg`.
- **Open software**: Python scripts, an 8-channel dataset and a Colab manual.

Typical applications

- Real-time EEG classification
- EEG combined with computer vision
- Neural decoding research
- Blink and jaw-clench control
- Edge-AI and robotics courses
- Neurofeedback prototypes

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	8 single-ended biosignal inputs
Extra inputs	Reference (SRB1, shared by all channels) and bias drive
Analog front end	Texas Instruments ADS1299
Resolution	24-bit delta-sigma, all channels sampled at the same moment
Sample rate	250 SPS set by PiEEG software (ADS1299 supports 250 SPS to 16 kSPS)
Gain	1 set by PiEEG software (ADS1299 supports 1, 2, 4, 6, 8, 12, 24)
Reference voltage	4.5 V internal; 1 LSB ≈ 0.268 µV at gain 1
Signals	EEG, EMG, ECG
HOST INTERFACE	
Host	NVIDIA Jetson Nano Developer Kit
Connector	Elevated 2 × 20 socket, 2.54 mm, onto the Jetson 40-pin header
Bus	SPI1 (`/dev/spidev0.0`) · mode 1 · 600 kHz · DRDY on header pin 7 (active low)

For research and education only. Not a medical device. Specifications subject to change.

Data frame	27 bytes per sample: 3-byte status + 8 × 3-byte channels
Indicators	LED D1 power · LED D2 connection
ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connector	1 × 10 header, 2.54 mm, marked "1-8 Elec": E1–E8, Biasout, Ref
POWER & SAFETY	
Supply	5 V through the Jetson Nano, from a 5 V / 3 A power bank (≤ 10,000 mAh) only
Regulation	Ultra-low-noise regulators on the shield
Isolation	None. Battery operation of the whole setup is required
Classification	Research and education tool. Not a medical device
MECHANICAL	
PCB	79.0 × 100.0 mm L-shape, 2-layer FR-4
Mounting	3 × Ø2.70 mm holes (M2.5)
Shipping weight	100 g (packed)
SOFTWARE	
PiEEG Server	`pieeg-server --device jneeg` — dashboard :1617, WebSocket :1616, CSV, LSL, OSC (Python 3.10+)
Python scripts	`1.Graph_real_time.py`, `2.Save_data.py` — run on the stock JetPack Python

03 BLOCK DIAGRAM

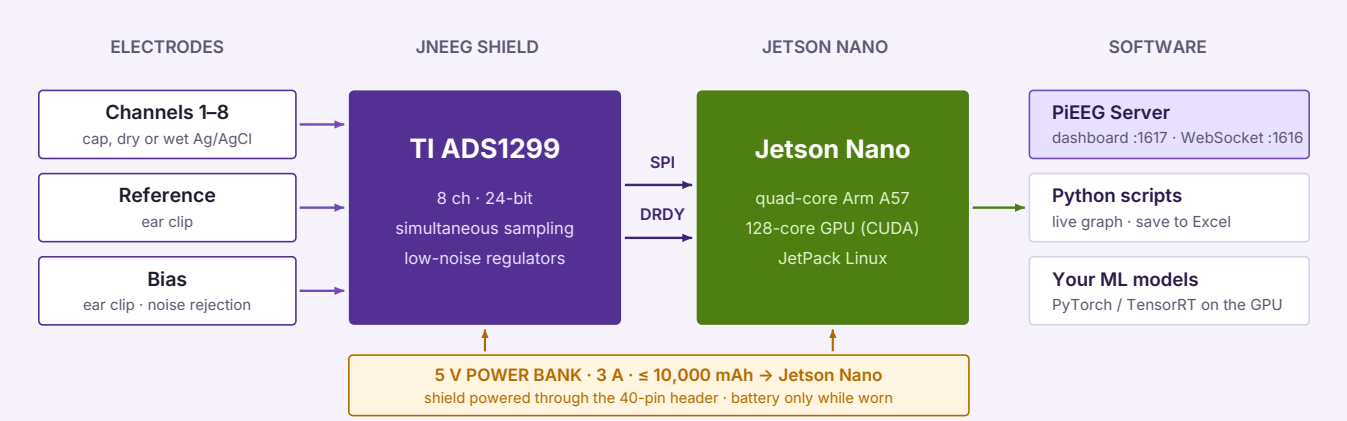


Figure 1 — Signal path. The ADS1299 digitises all channels together; the Jetson reads each frame over SPI when DRDY goes low and can process it on the GPU.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms: delta, theta, alpha, beta, gamma	Close the eyes and watch alpha (8–13 Hz) rise on O1/O2
EMG	Muscle activity	Clench the jaw or a fist
ECG	Heart activity	Electrodes on the chest or wrists

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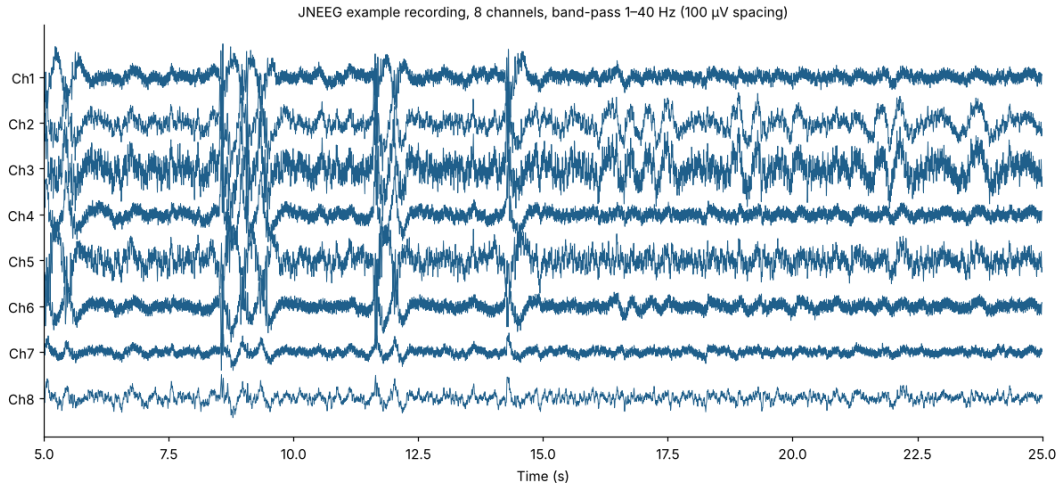


Figure 2 — 20 s of the JNEEG example dataset, 8 channels, band-pass 1–40 Hz.

05 PIEEG FAMILY

	JNEEG	PiEEG-8	ardEEG
Channels	**8**	8	8
Host	**Jetson Nano**	Raspberry Pi 4 / 5	Arduino UNO R4 WiFi
GPU	**Yes, 128 CUDA cores**	No	No
PiEEG Server	**--device jneeg**	--device pieeg8	—
List price	**USD 350**	USD 350	USD 350

06 ORDERING INFORMATION

Product	JNEEG — 8-channel EEG shield for NVIDIA Jetson Nano
SKU	CQA240229JN
Price	USD 350 (list)
Purchase	Elecrow (official manufacturing partner) — elecrow.com/jneeg.html · pieeg.com/jneeg

Included	
1 × JNEEG shield	ADS1299, regulators, 40-pin socket, electrode header

Not included — required	
Jetson Nano Dev Kit	microSD 32 GB+ with JetPack
5 V / 3 A power bank	Battery only
Electrodes and cap	8 Ag/AgCl + 2 ear clips
Monitor, keyboard	Battery-powered, wired

WARNING Battery only.

JNEEG has no isolation. The Jetson, the shield and everything cabled to them must run from a 5 V power bank while electrodes are on a person — never the Jetson's wall supply. JNEEG is not a medical device.

07 RELATED DOCUMENTS

JN-MD-002 Mechanical & Dimensional Drawing	**JN-PCB-003** PCB & Manufacturing Specification	**JN-PIN-004** Connector Pinout & Electrode Guide
JN-QSG-005 Quick Start & Safety Guide	**JN-SW-006** Data Interface & Software Guide	

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